

This is an excellent place to introduce **React Router DOM**. Instead of teaching each hook separately, build **one real-world multi-page React application**. Students will understand routing much faster.

React Router DOM Navigation

Slide 1. Learning Objectives

React Router DOM

Today You Will Learn

- What is React Router?
- BrowserRouter
- Routes
- Route
- Link
- NavLink
- Navigate
- useNavigate()
- useParams()
- Nested Routes
- 404 Page

Project

Build a Responsive Company Website with Navigation.

Learning Outcome

Students will build a professional multi-page React application similar to real company websites.

Slide 2. What is React Router?

React Router allows navigation between different pages without refreshing the browser.

Traditional Website

Home.html

About.html

Contact.html

Every page reloads.

React Router

Home

About

Services

Products

Contact

Same application.

Only content changes.

Benefits

- Fast Navigation
 - Better User Experience
 - Single Page Application (SPA)
 - Industry Standard
-

Slide 3. Installation & Folder Structure

Install

`npm install react-router-dom`

Folder Structure

src

components

Header.jsx

pages

Home.jsx

About.jsx

Products.jsx

ProductDetails.jsx

Contact.jsx

Dashboard.jsx

Profile.jsx

Settings.jsx

NotFound.jsx

App.jsx
main.jsx

BrowserRouter

main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

import { BrowserRouter } from "react-router-dom";

ReactDOM.createRoot(
  document.getElementById("root")
).render(
  <BrowserRouter>

  <App/>

</BrowserRouter>

);
```

Slide 4. Routes and Route

App.jsx

```
import { Routes, Route } from "react-router-dom";

import Home from "./pages/Home";
import About from "./pages/About";
import Contact from "./pages/Contact";

function App(){

  return(

    <>

    <Routes>

    <Route path="/" element={<Home/>}/>

    <Route path="/about" element={<About/>}/>

    <Route path="/contact" element={<Contact/>}/>

    </Routes>

    </>

  )
}
```

```
);  
  
}  
  
export default App;
```

URL

```
/  
  
/  
  
/about  
  
/contact
```

Slide 5. Bootstrap Responsive Header using Link & NavLink

Install Bootstrap

```
npm install bootstrap  
import "bootstrap/dist/css/bootstrap.min.css";
```

Header.jsx

```
import { Link, NavLink } from "react-router-dom";  
  
function Header(){  
  
  return(  
  
    <nav className="navbar navbar-expand-lg navbar-dark bg-dark">  
  
      <div className="container">  
  
        <Link  
          className="navbar-brand"  
          to="/"  
        >  
          TechDairy  
        </Link>  
  
        <button  
          className="navbar-toggler"  
          data-bs-toggle="collapse"  
          data-bs-target="#menu"  
        >  
  
        <span className="navbar-toggler-icon"></span>
```

```
</button>
```

```
<div  
  className="collapse navbar-collapse"  
  id="menu"  
>
```

```
<ul className="navbar-nav ms-auto">
```

```
<li className="nav-item">
```

```
<NavLink  
  to="/"   
  className="nav-link"  
>  
  Home  
</NavLink>
```

```
</li>
```

```
<li>
```

```
<NavLink  
  to="/about"   
  className="nav-link"  
>  
  About  
</NavLink>
```

```
</li>
```

```
<li>
```

```
<NavLink  
  to="/products"   
  className="nav-link"  
>  
  Products  
</NavLink>
```

```
</li>
```

```
<li>
```

```
<NavLink  
  to="/contact"   
  className="nav-link"  
>  
  Contact  
</NavLink>
```

```
</li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
</nav>

);

}

export default Header;
```

Difference

Link

Navigates between pages.

NavLink

Automatically adds an active class to the current page.

Slide 6. Navigate & useNavigate()

Navigate Component

Redirect users automatically.

```
import { Navigate } from "react-router-dom";

function Login(){

  const isLogin=true;

  return(

    isLogin

    ?

    <Navigate to="/dashboard"/>

    :

    <h2>Please Login</h2>

  );

}
```

useNavigate()

Navigate after a button click.

```
import { useNavigate } from "react-router-dom";

function Home(){

  const navigate=useNavigate();

  return(

    <button

      className="btn btn-primary"

      onClick={()=>navigate("/products")}

    >

      View Products

    </button>

  );

}
```

Real World

- Login
- Logout
- Checkout
- Payment Success

Slide 7. useParams()

Dynamic Routing

App.jsx

```
<Route

  path="/products/:id"

  element={<ProductDetails/>}

/>
```

ProductDetails.jsx

```
import { useParams } from "react-router-dom";
```

```
function ProductDetails(){
```

```
  const {id}=useParams();
```

```
  return(
```

```
    <h2>
```

```
    Product Id : {id}
```

```
  </h2>
```

```
);
```

```
}
```

URL

/products/5

Output

Product Id : 5

Real World

- Product Details
- Student Details
- Employee Profile
- Blog Details

Slide 8. Nested Routing

Dashboard.jsx

```
import { Outlet, Link } from "react-router-dom";
```

```
function Dashboard(){
```

```
  return(
```

```
    <>
```

```
    <h2>Dashboard</h2>
```

```
    <Link to="profile">
```

```
    Profile
```

```
  </Link>
```



```
<br/>  
  
<Link to="settings">
```

Settings

```
</Link>
```

```
<hr/>
```

```
<Outlet/>
```

```
</>
```

```
);
```

```
}
```

Routes

```
<Route  
  path="/dashboard"  
  element={<Dashboard/>}  
>
```

```
<Route  
  path="profile"  
  element={<Profile/>}  
</>
```

```
<Route  
  path="settings"  
  element={<Settings/>}  
</>
```

```
</Route>
```

URL

/dashboard/profile

/dashboard/settings

Slide 9. Mini Project

Responsive Company Website

Build

Bootstrap Navbar



Home



About



Products



Contact



Product Details



Dashboard



Profile



Settings

Concepts Used

- BrowserRouter
 - Routes
 - Route
 - Link
 - NavLink
 - useNavigate()
 - useParams()
 - Nested Routing
-

Slide 10. Homework

Company Website with Routing

Pages

- Home
- About
- Services
- Products
- Contact
- Dashboard
- Profile
- Settings
- 404 Page

Requirements

- Responsive Bootstrap Navbar
- Active Navigation using NavLink
- Dynamic Product Details using useParams()
- Dashboard with Nested Routes
- Redirect to Home using Navigate
- Button Navigation using useNavigate()
- Custom 404 Page for invalid URLs

Submission Checklist

🔍 BrowserRouter

🔍 Routes

🔍 Route

🔍 Link

🔍 NavLink

🔍 Navigate

🔍 useNavigate()

🔍 useParams()

🔍 Nested Routing

🔍 404 Page

Expected Outcome

Students will build a complete multi-page React application with a responsive Bootstrap navigation bar, dynamic routing, nested routes, protected navigation, and a custom 404 page. This mirrors the routing structure commonly used in production React applications such as e-commerce platforms, admin dashboards, and SaaS products.